

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – Technical Presentation

|                         |                                                                                                                                                          |                      |                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|
| <b>Course Code:</b>     | DSA0306                                                                                                                                                  | <b>Course Title:</b> | Natural Language Processing                |
| <b>CO Assessed:</b>     | CO1 – Imitation (L1)                                                                                                                                     | <b>Assessment:</b>   | Assessment Tool 2 – Technical Presentation |
| <b>Weightage:</b>       | 30% of CIA                                                                                                                                               |                      |                                            |
| <b>CO1 Statement:</b>   | Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3) |                      |                                            |
| <b>Student Name:</b>    | Y Sravan Kumar                                                                                                                                           | <b>Reg. No.:</b>     | 192472289                                  |
| <b>Section / Batch:</b> | CSE AI                                                                                                                                                   | <b>Date:</b>         | 16-07-2026                                 |

### Code of Conduct

*I Y. Sravan Kumar (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_\_ Y. Sravan Kumar \_\_\_\_\_

### Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

| Section / Topic                                                                 | Focus                                           | Q Nos |
|---------------------------------------------------------------------------------|-------------------------------------------------|-------|
| Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | NLP Models, Language Processing Evolution       | TP-01 |
| Comparative Study of NLP Models and Algorithms                                  | Model Analysis, Algorithm Comparison            | TP-02 |
| Regular Expressions for Real-World Text Processing                              | Pattern Matching, Information Extraction        | TP-03 |
| Designing Efficient Regular Expression Patterns for Information Extraction      | Text Mining, Data Extraction Techniques         | TP-04 |
| Finite State Automata in Natural Language Processing                            | State Transitions, Language Recognition         | TP-05 |
| Building a Finite State Automaton for Token Recognition                         | Lexical Analysis, Token Validation              | TP-06 |
| Morphology and Word Formation in Natural Languages                              | Morpheme Analysis, Word Structure Processing    | TP-07 |
| Inflectional Morphology: Understanding Grammatical Variations                   | Grammatical Analysis, Word Normalization        | TP-08 |
| Derivational Morphology and Vocabulary Expansion                                | Word Generation, Semantic Transformation        | TP-09 |
| Comparative Analysis of Inflectional and Derivational Morphology                | Morphological Classification, Language Analysis | TP-10 |

|                                                                               |                                                |       |
|-------------------------------------------------------------------------------|------------------------------------------------|-------|
| Finite-State Morphological Parsing Techniques                                 | Morphological Parsing, Finite-State Processing | TP-11 |
| Morphological Analysis for Indian Languages                                   | Language-Specific Morphology, NLP Challenges   | TP-12 |
| Porter Stemmer: Design, Working Mechanism, and Applications                   | Stemming Algorithms, Text Preprocessing        | TP-13 |
| Stemming vs Morphological Parsing: A Comparative Study                        | Word Reduction, Morphological Processing       | TP-14 |
| Integrated NLP Pipeline: From Regular Expressions to Morphological Processing | End-to-End Language Processing Pipeline        | TP-15 |

## Annexure A – Technical Presentation

| Team No. | Technical Presentation Title                                                    | Description                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.              |
| 2        | Comparative Study of NLP Models and Algorithms                                  | Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.                             |
| 3        | Regular Expressions for Real-World Text Processing                              | Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.                 |
| 4        | Designing Efficient Regular Expression Patterns for Information Extraction      | Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.                    |
| 5        | Finite State Automata in Natural Language Processing                            | Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.                              |
| 6        | Building a Finite State Automaton for Token Recognition                         | Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.                                    |
| 7        | Morphology and Word Formation in Natural Languages                              | Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.                                             |
| 8        | Inflectional Morphology: Understanding Grammatical Variations                   | Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.                                               |
| 9        | Derivational Morphology and Vocabulary Expansion                                | Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.                                    |
| 10       | Comparative Analysis of Inflectional and Derivational Morphology                | Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.                                                   |
| 11       | Finite-State Morphological Parsing Techniques                                   | Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.                                   |
| 12       | Morphological Analysis for Indian Languages                                     | Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.        |
| 13       | Porter Stemmer: Design, Working Mechanism, and Applications                     | Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.                                      |
| 14       | Stemming vs Morphological Parsing: A Comparative Study                          | Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.                                         |
| 15       | Integrated NLP Pipeline: From Regular Expressions to Morphological Processing   | Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing. |

## Rubrics for Calculation

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10        | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness    |

| Team No. / Criteria Level | Technical Content Accuracy | Problem Identification | Use of Tools | Communication | Professionalism | Total % |
|---------------------------|----------------------------|------------------------|--------------|---------------|-----------------|---------|
| 7                         |                            |                        |              |               |                 |         |

| Faculty In-charge                 | Course Coordinator | Program Director |
|-----------------------------------|--------------------|------------------|
|                                   |                    |                  |
| Signature :Dr Kumaragurubaran____ | Signature: _____   | Signature: _____ |
| Date: ____16-07-2026_____         | Date: _____        | Date: _____      |



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# **Morphology and Word Formation in Natural Languages**

## **DSA0306-Natural Language Processing**

**Guided By :**

**Dr. Kumaragurubaran**

**Presented By :**

**Y . Sravan Kumar (192472289)**



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# Introduction

## What is Morphology?

- Morphology is the branch of linguistics that studies the **internal structure of words** and how new words are formed using the smallest meaningful units called **morphemes**.

## Key Points

- Studies word formation.
- Analyzes word structure.
- Identifies morphemes.
- Helps understand language rules.
- Essential in Natural Language Processing (NLP).

## Example

Unhappiness

➤ Un + Happy + Ness

1. **Un** → Prefix

2. **Happy** → Root Word

3. **Ness** → Suffix



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# Morphemes

## What is a Morpheme?

- A morpheme is the smallest meaningful unit in a language.

### Types of Morphemes

#### Free Morphemes

Can stand alone.

Examples

Book

Car

Happy

Play

#### Bound Morphemes

Cannot stand alone.

Examples

un-

-ed

-ing

-s

-er

# Word Formation Process

## How Words are Formed

- Root word  $\longrightarrow$  Prefix + Root word + Suffix  $\longrightarrow$  New Word

## Examples

| Root   | Prefix | Suffix | New Word      |
|--------|--------|--------|---------------|
| Happy  | Un     | Ness   | Unhappiness   |
| Teach  | —      | Er     | Teacher       |
| Nation | Inter  | Al     | International |
| Read   | Re     | —      | Reread        |





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# Types of Morphology

- **Inflectional Morphology**

- Changes grammatical form.
- Meaning remains the same.
- Examples
  - Play → Played
  - Cat → Cats
  - Tall → Taller

- **Derivational Morphology**

- Creates a new word.
- Meaning may change.
- Examples
  - Happy → Happiness
  - Teach → Teacher
  - Kind → Unkind



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# Morphological Analysis in NLP

Morphological analysis breaks words into meaningful parts.

Word



Tokenization



Morpheme Identification



Root Extraction



Word Analysis



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# Importance of Morphology in NLP

## Why Morphology is Important

- Improves machine translation.
- Supports spell checking.
- Enhances grammar checking.
- Helps information retrieval.
- Improves search engines.
- Supports speech recognition.
- Enables text summarization.
- Assists sentiment analysis.
- Helps chatbots understand user input.



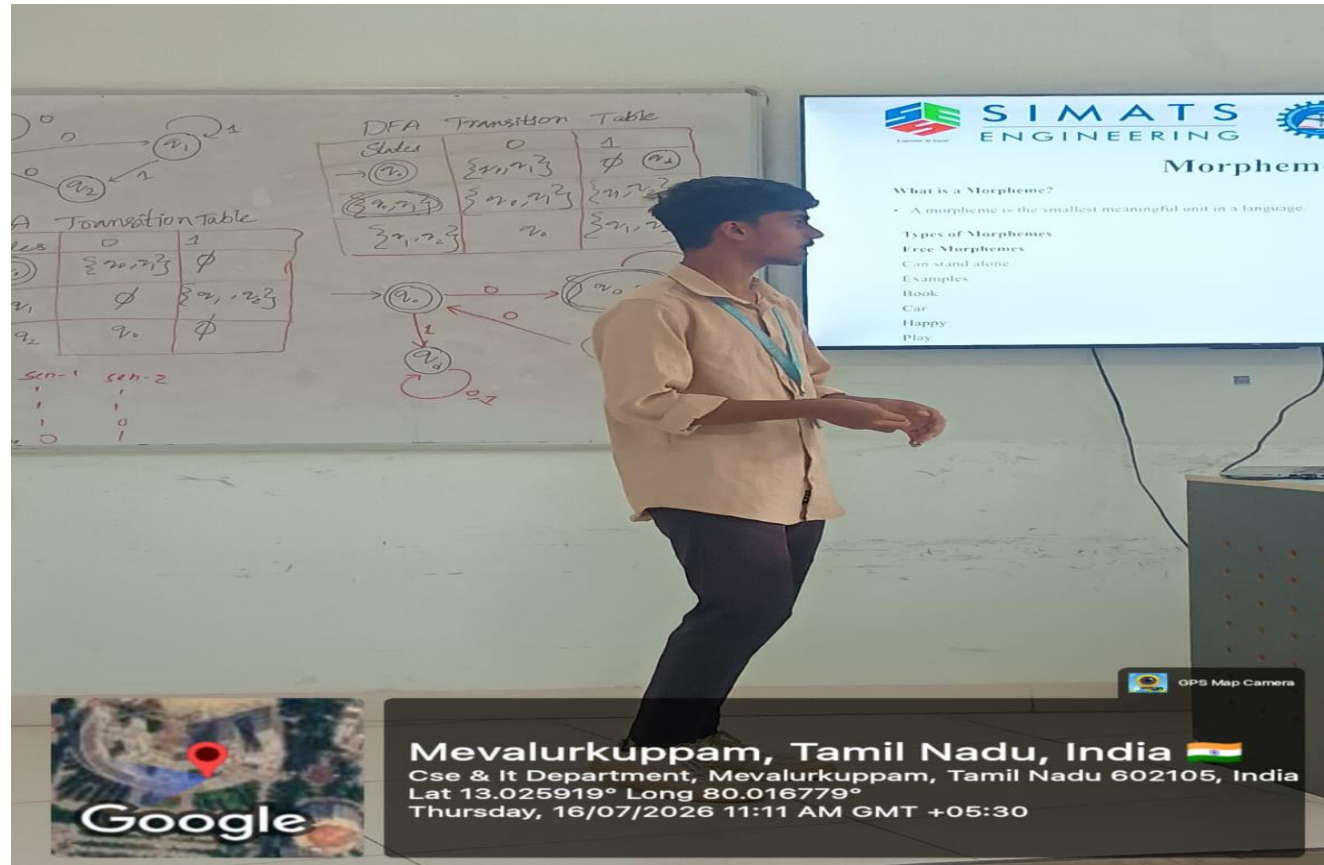
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# Conclusion

- Morphology studies the internal structure of words.
- Morphemes are the basic building blocks of language.
- Word formation involves prefixes, suffixes, and root words.
- Morphological analysis improves language understanding.
- It plays a vital role in modern NLP applications.
- Morphology enhances the performance of AI-based language systems.



**DFA Transition Table**

| States            | 0              | 1              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_1, q_2\}$ | $\phi$         |
| $\{q_1, q_2\}$    | $\{q_0, q_2\}$ | $\{q_1, q_2\}$ |
| $\{q_1, q_2\}$    | $q_0$          | $\{q_1, q_2\}$ |

**A Transition Table**

|       | 0              | 1              |
|-------|----------------|----------------|
| $q_0$ | $\{q_1, q_2\}$ | $\phi$         |
| $q_1$ | $\phi$         | $\{q_1, q_2\}$ |
| $q_2$ | $q_0$          | $\phi$         |

**Morphemes**

**What is a Morpheme?**

- A morpheme is the smallest meaningful unit in a language.

**Types of Morphemes**

**Free Morphemes**

- Can stand alone
- Examples
- Book
- Car
- Happy
- Play

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Thank You